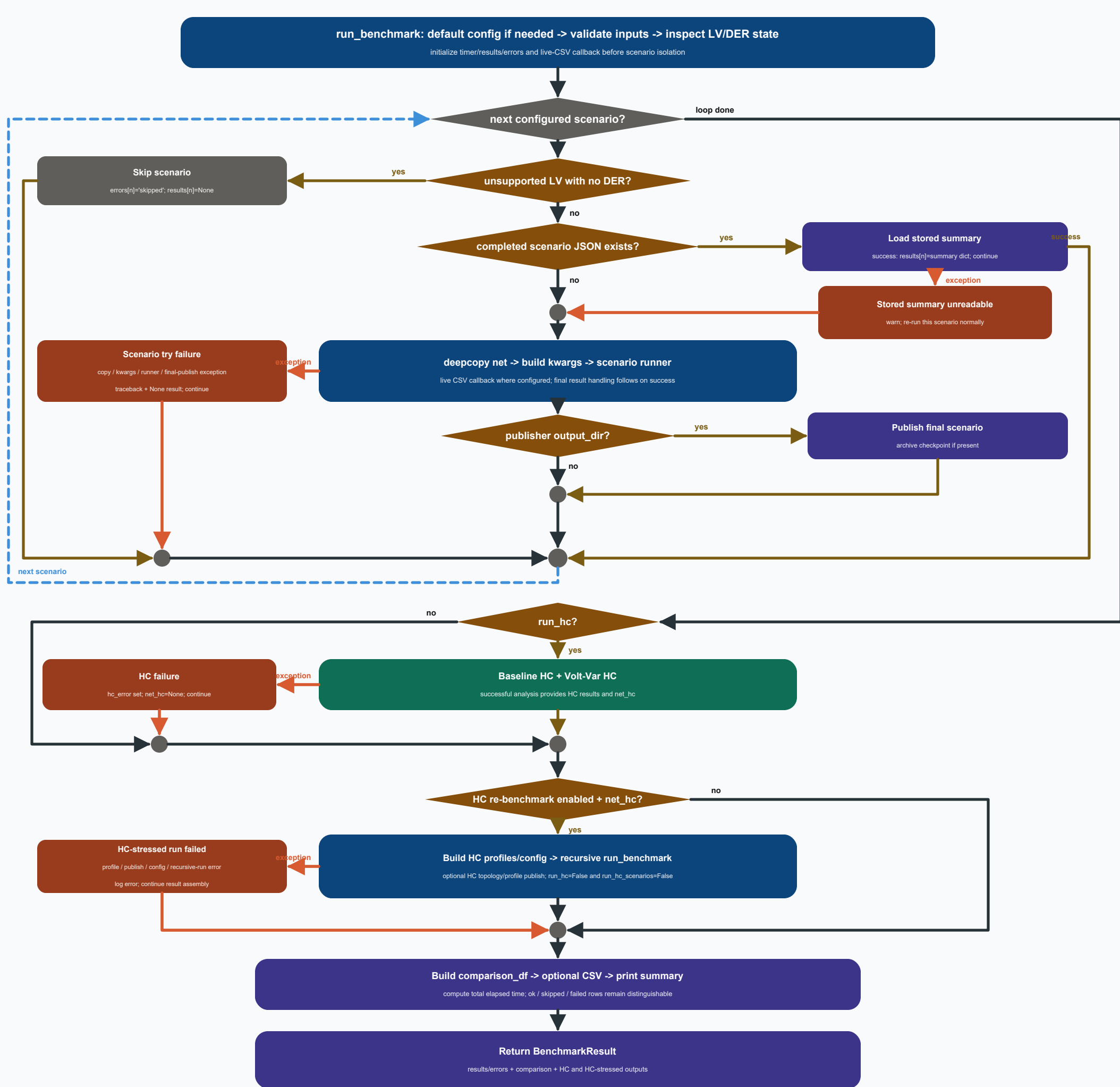




## Scenario isolation and dispatch

Validation runs once before the loop and checks profiles, scenario IDs, hardware settings and HC prerequisites.

The runner also records whether the network is LV and whether an in-service PV/wind DER fleet already exists.

Configured scenarios run in ascending numeric order; each receives its own `copy.deepcopy(net)`.

`_build_kwargs` injects only the hardware, coordination, OPF, publishing and checkpoint options supported by that runner.

An exception during isolated execution or final publication is stored as a failed result; the next scenario still runs.

## Skip, resume and publication layers

Unsupported LV scenarios are skipped only when the LV network also has no controllable PV/wind DERs.

An existing final scenario JSON can be loaded as a completed summary and that scenario is not rerun.

If the stored JSON cannot be read, the runner logs a warning and reruns that scenario from scratch.

When `live_csv_path` is configured, a callback rewrites a partial benchmark CSV during scenario execution.

After success, the final scenario JSON is published and any matching checkpoint stream is archived as `.completed`.

Skip, stored-summary success, normal success and failure all rejoin at the next-scenario loop boundary.

## Hosting capacity and result assembly

After the scenario loop, optional HC evaluates baseline and Volt-Var hosting capacity on the untouched original network.

HC failure is stored in `hc_error`, sets `net_hc=None` and does not abort the outer benchmark.

If HC-stressed scenarios are enabled and `net_hc` exists, `profile_factory` builds profiles for the stressed network.

The recursive config disables `run_hc` and `run_hc_scenarios` so a second HC study cannot recurse indefinitely.

HC-stressed setup/run failure is logged; comparison assembly still proceeds.

Finally, the runner builds `comparison_df`, optionally writes CSV, prints the summary and returns `BenchmarkResult`.